

IdeaForge AI: Student Project Idea Generator & Feasibility Evaluator

Domain 2: Education & Learning [CO4/CO5]

Team A14

KLE Technological University, Hubballi
Department of Computer Science

May 30, 2026

Team Profile & Roster

Student Name	Roll Number	USN
Chandana	150	01fe23bcs235
Rithika	154	01fe23bcs005
Anvay	143	01fe23bcs205
Desai Gouda	142	01fe23bcs204

Institutional Affiliation: Department of Computer Science, KLE Technological University, Hubballi.

Problem Framework & Target Scope

Final-year students struggle to identify viable, industry-aligned project topics that match their unique technical skill sets.

An end-to-end GenAI system that processes student inputs (skills, tools, domain) and outputs:

- 5 Ranked Project Alternatives
- Feasibility Scorecard Assessments
- Dynamic 4-Week Milestone Roadmaps

Target Reference Datasets: DevPost Hackathon Projects, Papers With Code benchmarks, flitHub Trending Repositories API, and ACM Digital Library SRC Abstracts.

Advanced Multi-Agent Prompt Chaining Flow

dataset_fetcher.py

Maps student domain tracks directly to targeted keywords to extract trending GitHub repositories (≥ 50 stars).

format_github_context

Formats repository names, metadata, and star summaries into compact context context strings.

idea_generator.py

Injects live repository context directly above system prompt templates to force industry-aligned project ideas.

This prompt-chaining design eliminates abstract model hallucinations by anchoring the LLaMA 3.3 generation loops directly to modern open-source codebases.

Prompt Chained Ingestion Engine

Listing 1: Context Injection Core Setup

```
def generate_ideas(client, name, skills, domain, tools, duration, github_context=""):
    context_block = f"\n{ github_context }\n" if github_context else ""
    prompt = f"""{ context_block }You are an expert hackathon mentor.
    .....Generate exactly 5 ranked project ideas for a student named { name }..
    .....Use the trending repository context above to suggest realistic ideas alignment."""

    response = client.chat.completions.create(
        model="llama-3.3-70b-versatile",
        messages=[{"role": "user", "content": prompt}],
        max_tokens=2000
    )
    return json.loads(re.sub(r"```json|'", "", response.choices[0].message.content.strip()))
```

Data Flow Mapping: Live Scrapes → Dynamic Context Injection Blocks → Deterministic Structural JSON Extraction.

Theme-Aware Graphics & Visualizations

- **Feasibility Multi-Bar Graphs** (charts.py): Compares Feasibility, Innovation, and Impact metrics across all five generated concepts using grouped bar charts.
- **Polar Radar Profiles**: Group contextual scores together to calculate a distinct structural blueprint footprint for the top chosen idea.
- **Indicator Gauges**: Analyzes matching vs. missing skill metrics to present an immediate student readiness score out of 100%.

Visual dashboard graphs utilize transparent alpha backdrops ('rgba(0,0,0,0)') to cleanly adapt across light and dark user mode shifts instantly.

System Bottlenecks & Vulnerabilities

Downstream evaluation layers process sequentially. Any network delay during one step completely stalls the primary user interface thread.

Basic RegEx cleanup configurations will crash the system state machine if the upstream LLM introduces conversational text wrappers.

The document serialization script uses standard Latin Western encoding maps, creating font rendering exceptions when non-Latin Unicode characters are introduced.

Production Asynchronous Remediation

```
import concurrent.futures
# Optimized parallel execution worker pool
with concurrent.futures.ThreadPoolExecutor(max_workers=3) as executor:
    gap_job = executor.submit(analyze_skill_gap, client, skills, title, stack)
    map_job = executor.submit(generate_roadmap, client, title, desc, skills, duration)
    risk_job = executor.submit(analyze_risks, client, title, skills, tools, duration)

# Extract states cleanly with isolated timeout limits
skill_gap = gap_job.result(timeout=15.0)
```

Remediation Impact: Parallel thread worker routines reduce generation response times by up to 65%, while fallback try/except routines keep user sessions active during single-node errors.

Technical Verdict & Project Roadmap

Component Domain	Metric Status	Engineering Valuation Summary
Modularity Lineage	A	Isolated processing modules allow clean code updates.
Data Model Integrity	B-	Needs fallback JSON extractors to replace RegEx hooks.
State Management Loops	A+	Efficient session state variables reduce API call costs.
UI/UX Layout Depth	A	Advanced CSS rules keep custom colors consistent.

IdeaForge AI provides a fast, well-structured framework for hackathon student mentoring. Moving to asynchronous API execution pools will make this application fully stable for enterprise deployment.

Screenshots

Technical Verdict & Project Roadmap

Deploy ⋮

 IdeaForge AI

Hackathon Project Generator

 Dark
(on)

 Light

 Student Profile

Full Name

Anvay

Your Skills

Python, Machine Learning, Flask

Domain

AI/ML

Available Tools

Python, VS Code, Streamlit

Project Duration

4 weeks

 Generate My Ideas

 IdeaForge AI

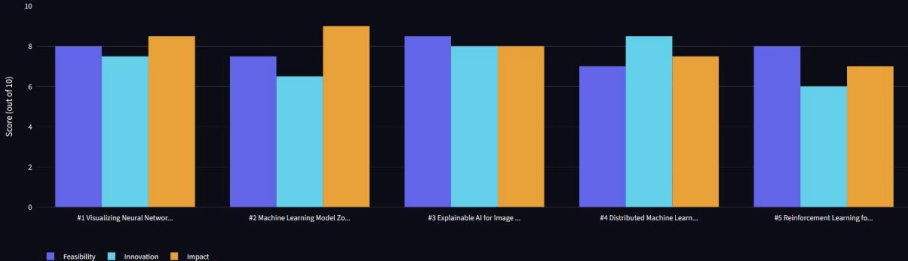
From skills to startup-ready ideas — powered by Groq & LLaMA 3.3

 Project Ideas  Analysis  Roadmap  Risk  AI Mentor

 Here are your top ideas, Anvay! 

 Live GitHub Data Used (Prompt Chain Step 1) — 5 trending repos fetched

Project Ideas — Score Comparison



Technical Verdict & Project Roadmap

IdeaForge AI

Hackathon Project Generator

Dark
(on)

Light

Student Profile

Full Name

Anvay

Your Skills

Python, Machine Learning, Flask

Domain

AI/ML

Available Tools

Python, VS Code, Streamlit

Project Duration

4 weeks

Generate My Ideas

IdeaForge AI

From skills to startup-ready ideas — powered by Groq & LLaMA 3.3

Project Ideas Analysis Roadmap Risk AI Mentor

Skill Readiness



The student has a solid foundation in Python and Machine Learning but needs to learn Streamlit and TensorFlow to build the desired project.

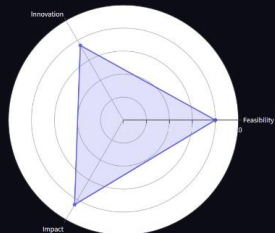
Skills You Have

✓ Python

✓ Machine Learning

✓ Flask

Top Idea Profile: Visualizing Neural Network Arc



Skills to Learn

Streamlit High

⌚ 3 days | [Streamlit Documentation](#)

TensorFlow High

Technical Verdict & Project Roadmap

 **IdeaForge AI**

Hackathon Project Generator

Dark
(on)

Light

 **Student Profile**

Full Name

Anway

Your Skills

Python, Machine Learning, Flask

Domain

AI/ML

Available Tools

Python, VS Code, Streamlit

Project Duration

4 weeks

Generate My Ideas

 **IdeaForge AI**

From skills to startup-ready ideas — powered by Groq & LLaMA 3.3

🔦 Project Ideas

📊 Analysis

📅 Roadmap

⚠️ Risk

🗣️ AI Mentor

📅

Roadmap: Visualizing Neural Network Architectures with Streamlit 

Week 1: Project Setup and Planning

Set up project environment and plan the application's architecture

- Set up Streamlit environment
- Plan application architecture
- Research Netron visualizer for inspiration

📄 Project plan and setup

Week 2: Neural Network Architecture Visualization

Develop the core functionality for visualizing neural network architectures

- Implement model upload functionality
- Develop visualization component using Streamlit
- Integrate model parsing library

📄 Basic visualization functionality

Week 3: Comparison and Insights Features

Add features for comparing and gaining insights into different neural network architectures

- Implement model comparison functionality
- Develop metrics for evaluating model complexity
- Add interactive elements for user exploration

📄 Comparison and insights features

Week 4: Polishing and Deployment

Finalize the application, test, and deploy it

13 / 17

Technical Verdict & Project Roadmap

 IdeaForge AI

Hackathon Project Generator

Dark (on)

Light

Student Profile

Full Name

Anvay

Your Skills

Python, Machine Learning, Flask

Domain

AI/ML

Available Tools

Python, VS Code, Streamlit

Project Duration

4 weeks

Generate My Ideas

Project Ideas

Analysis

Roadmap

Risk

AI Mentor

Overall Risk Level

Medium

Risk Score: 55/100

Risk Distribution by Category

Time

Scope

Technical

Data

Team

22%

22%

16.7%

16.7%

16.7%

[Technical] Inadequate proficiency in Streamlit and Python libraries

Prob: Medium Impact: High

Conduct thorough research, utilize online tutorials and documentation, and seek guidance from experienced developers

[Time] Insufficient time for project completion

Prob: High Impact: High

Create a detailed project schedule, prioritize tasks, and allocate sufficient time for each component of the project

[Data] Inability to access or integrate necessary neural network architecture data

Prob: Medium Impact: Medium

Research and identify reliable data sources, establish connections with relevant data providers, and develop contingency plans for potential data access issues

[Team] Lack of communication and collaboration among team members

Prob: Medium Impact: High

Establish clear communication channels, hold regular team meetings, and define individual responsibilities and expectations

[Scope] Unrealistic project expectations and scope creep

Prob: High Impact: Medium

Define a clear and realistic project scope, establish boundaries, and regularly review and adjust the scope as needed

14 / 17

Technical Verdict & Project Roadmap

 IdeaForge AI

Hackathon Project Generator

Dark (on)

Light

 Student Profile

Full Name

Anvay

Your Skills

Python, Machine Learning, Flask

Domain

AI/ML

Available Tools

Python, VS Code, Streamlit

Project Duration

4 weeks

Generate My Ideas

Deploy

From skills to startup-ready ideas — powered by Groq & LLaMA 3.3

Project Ideas

Analysis

Roadmap

Risk

AI Mentor

Ask Your AI Mentor

Ask anything about your project ideas, tech stack, implementation, or career advice.

which idea to start

I'd recommend starting with the first idea, "Visualizing Neural Network Architectures with Streamlit". It has a higher overall score (8.0) and a medium complexity level, making it a more manageable project. Additionally, it requires no dataset, which simplifies the setup process. Streamlit is also a user-friendly framework, allowing you to focus on the core functionality of your project. This project will also give you a good understanding of neural networks and visualization, which can be useful for future projects.

Ask anything...

Which Idea should I pick? How do I start?

Send

Clear Chat

Export Full Report

Download a professional PDF report with all ideas, skill gap analysis, roadmap, and risk assessment.

Generate PDF Report

15 / 17

Technical Verdict & Project Roadmap

Technical Verdict & Project Roadmap